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Mortality through the early life-history of fish: What can we learn from European plaice (*Pleuronectes platessa* L.)?

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ABSTRACT

It is difficult to estimate natural mortality for many marine fish populations, especially during the transition period from larvae to juveniles, because the appropriate data are scarce. Plaice (*Pleuronectes platessa*) is an exception since it has been studied extensively. The study of mortality rates in juveniles is made easier because the nursery grounds are inshore and generally less than 5 m deep. This contribution considers the factors affecting mortality rates for eggs and larvae, and for settlement and nursery ground phases. There are problems associated with estimating mortality rates, especially for juveniles because immigration into nursery areas at the beginning of the season and emigration of larger individuals off nursery grounds in the latter part of the season confound losses due to mortality. The shifts in mortality schedules and the causes through the early life history are investigated in relation to concepts such as 'nursery ground carrying capacity' and 'self-thinning'. Predation in the pelagic phase is probably density independent and a source of inter-annual variability in survival of early life-history stages. Predation mortality during the nursery ground phase is most likely density dependent. However, there is a need for further in-depth study, especially during the period of settlement.

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1. Introduction

The causes of mortality in fish populations vary considerably and the sources of mortality can be broadly separated into endogenous and exogenous factors. During the past century of fisheries and marine ecology research numerous hypotheses have been stated to explain the actions of these factors and to assign precedence to those factors which can reliably be quantified in order to predict interannual variations in survivorship. Growth is one factor that is inextricably linked to mortality or survival rates, often viewed through hypotheses on size-selective mortality (see Sogard, 1997 for a review). Endogenous factors tend to occur early in life e.g. egg and larvae stages (Houde, 2002), sometimes associated with changes in physiology or structure e.g. metamorphosis (Geffen et al., 2007). Developmental mortality is often caused by maternal or paternal influences through genetic abnormalities. As the early life history stage progresses exogenous factors become more prevalent such as physical factors (e.g. temperature, salinity or oxygen regimes), predation, starvation, and disease (Houde, 1987, 2002). Some of these factors are more explicitly density dependent. Physical transport mechanisms can carry individuals in the very early pelagic stages out of the optimal survival habitats, as suggested by Hjort (1914) in the hypothesis concerning the drift of young stages and then encapsulated in the Member-Vagrant Hypothesis (Sinclair, 1988). Mortality is high among larvae which do not reach a suitable nursery area, from which they can recruit back to the adult population and thus complete 'life history closure' (see Rijnsdorp et al., 1995). Thus, spatial factors are important, considering the location of spawning grounds and suitable nursery grounds and the concept of connectivity between the two (Sinclair, 1988).

In this review, we examine the patterns of early life history stage mortality using the European plaice (*Pleuronectes platessa*) as a model. We focus especially on the mortality processes during the later larval and early juvenile stages, since settlement and nursery ground stages have particular characteristics that influence mortality. Using a Paulik diagram approach (see Nash, 1998; Paulik, 1973), we briefly review the information about pelagic stage mortality (Quadrant 2), and then present detailed analyses/review of the processes occurring in the transition to nursery grounds (transition to Quadrant 3) and throughout the first year of life (Quadrant 3).

Plaice are useful as an example species because of the amount of data available on abundance and mortality rate calculations during the pelagic and nursery ground stages. Whilst there is a large amount of data on mortality rates of a number of juvenile flatfish on nursery grounds e.g. plaice, dab (*Limanda limanda*), sole (*Solea solea*) and turbot (*Scopthalmus maximus*), much of which is summarized in Iles and Beverton (1991), the majority of the available information is for plaice. Juvenile plaice populations have been studied at the extremes

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of its range: in Iceland (Hjörleifsson and Pálsson, 2001), the west coast of Ireland (Allen et al., 2008), the Swedish Kattegat (Modin and Pihl, 1994; Pihl, 1990; Wennhage et al., 2007). Nursery ground settlement patterns are also known for beaches in the Irish Sea (Al-Hossaini et al., 1989; Fox et al., 2007), the North Sea (Amara and Paul, 2003; Lockwood, 1974), and the coasts of Scotland (Steele and Edwards, 1970). Most importantly, there are three areas in particular that have supported long time series of plaice nursery ground studies: the Wadden Sea (van der Veer et al., 1990), the west coast of Scotland (Gibson et al., 1993) and Port Erin Bay, Isle of Man (Nash and Geffen, 1999, 2000; Nash et al., 2007). Most of the examples in this review are based on analyses of juvenile plaice surveyed through the past 30 years in these areas.

2. Development of plaice early life history mortality studies

Much of the early work on mortality rates concentrated on the pelagic egg and larvae stages in the life cycle, leading to the concepts of 'critical periods' (e.g. Hjort, 1914, 1926), 'match-mismatch' with feeding conditions (Cushing, 1975, 1990), and 'bigger is better' whereby size determines the mortality rate (see Leggett and Deblois, 1994) or 'stage duration' vulnerability whereby mortality rate may be relatively constant but the longer individuals remain within a stage (slower development times) the greater the numbers that are lost (Chambers and Legget, 1987; Houde, 1987). However, less research has been directed at detailed examinations of mortality during the juvenile phase of the life history, with a notable exception being a limited selection of flatfish species (e.g. Gibson, 2005). Flatfishes provide an interesting example in regard to nursery grounds in that the fish move from a three to two dimensional habitat when settling at the nurseries and thus invoke what Beverton (1995) referred to as the "concentration hypothesis" and the linking of spatial aspects with population size.

Mortality rates for early stages of plaice were first derived as generational mortality functions. Stage dependent mortality is not considered in such an analysis. For plaice in the North Sea, the proportion of survivors from egg to age 1, over the 1957 to 2006 year classes had a range of 13-fold (utilizing egg production supplied by F. Scott et al. Cefas, Lowestoft, *pers. comm.*). In the Irish Sea, survival to age 1 for the 1963 to 2004 year classes had a range of 8-fold (utilizing fecundity data from Kennedy et al. (2008) and population data from ICES (2008)). The annual variability is illustrated in Fig. 1. In the Barents Sea plaice population the survival from egg to age 4+ had a range of 400-fold over the 1972 to 1988 year classes (Kovtsova and Boitsov, 1995). Assuming a median spawning date of 20th January for the North Sea (based on data in van Damme et al. (2009)) and 10th March for the Irish Sea (based on data in Nash and Geffen (1999), Armstrong et al. (2001), and Fox et al. (2007)) gives an average generational instantaneous daily mortality (eggs to age 1) of 0.031 d^{-1} for the North Sea and 0.036 d^{-1} for the Irish Sea. Whilst survivorship was slightly higher (or mortality rates slightly lower) in the North Sea compared with the Irish Sea the two areas or populations were similar.

The generational mortality function provides a reasonable estimate of the numbers of individuals at age 1, but it will not necessarily estimate the numbers of individuals at intermediate time periods if there are stage dependent mortality rates (Fig. 2). Beverton and Iles (1992b) and Beverton (1995) estimated the population size of plaice in the North Sea from egg production, at hatching, metamorphosis/settlement on the nursery grounds, and through the nursery ground phase to 1st January i.e. beginning of age 1 (Fig. 3a). It is obvious that the mortality rates change through each phase of the life history and the degree of inter-annual variability in abundance at each 'transition' point or shift from one phase to the next was different. Abundance at hatching varied three-fold, and inter-annual variability in abundance at settlement increased to 200-fold. Through the nursery ground phase to the start of age 1 the variance decreased again (dampened) to approximately four-fold. A similar pattern was seen for Irish Sea plaice (Fig. 3b), although with a steady increase in variability throughout the pelagic phase.

To examine this variability in more detail it is possible to follow the annual development of abundance through the early life history. One method of doing this visually is by using a variation on the original diagrams presented by Paulik (1973), as used by Nash (1998), Nash and Dickey-Collas (2005), and Payne et al. (2009). The Paulik diagram developed in Nash (1998) has been reworked to include temperature modulated egg development rates (Fox et al., 2003), a temperature dependent egg mortality rate (Dickey-Collas et al., 2003), and updated to the 2006 year class (Fig. 4). There is a substantial increase in the interannual variability in survival between egg production and metamorphosis or arrival on the nursery ground (Table 1). Since the Paulik diagram quadrants are sequential, with a common axis between life stages, it is possible to utilize data that are collected at different scales e.g. at the population level or a single nursery ground. The patterns are self evident without the need for the

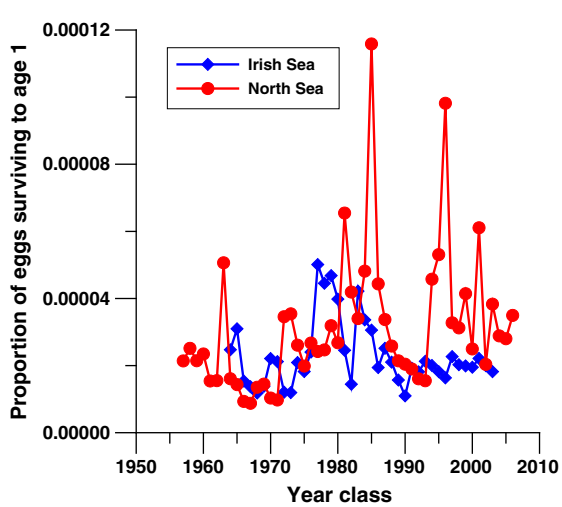

Fig. 1. Proportion surviving from egg to age 1 (generational mortality) for North Sea (circles) and Irish Sea (diamonds) plaice.

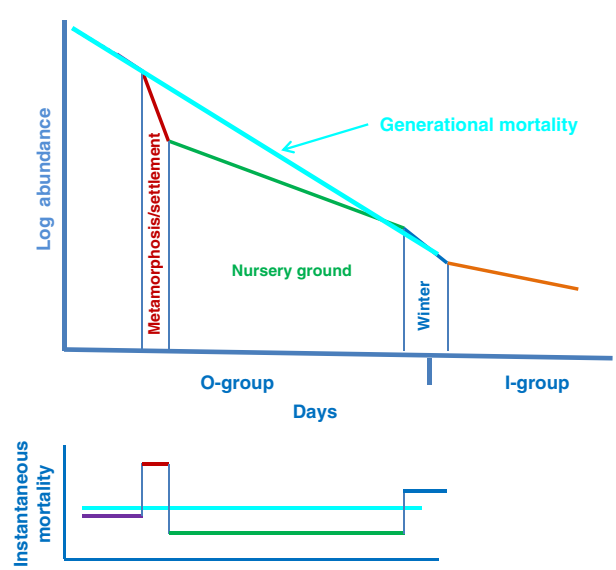

Fig. 2. Illustration of generational mortality and stage (pelagic, metamorphosis/settlement, nursery ground, over wintering) dependent mortality rates during the early life history phase of fishes. Example based on European plaice.

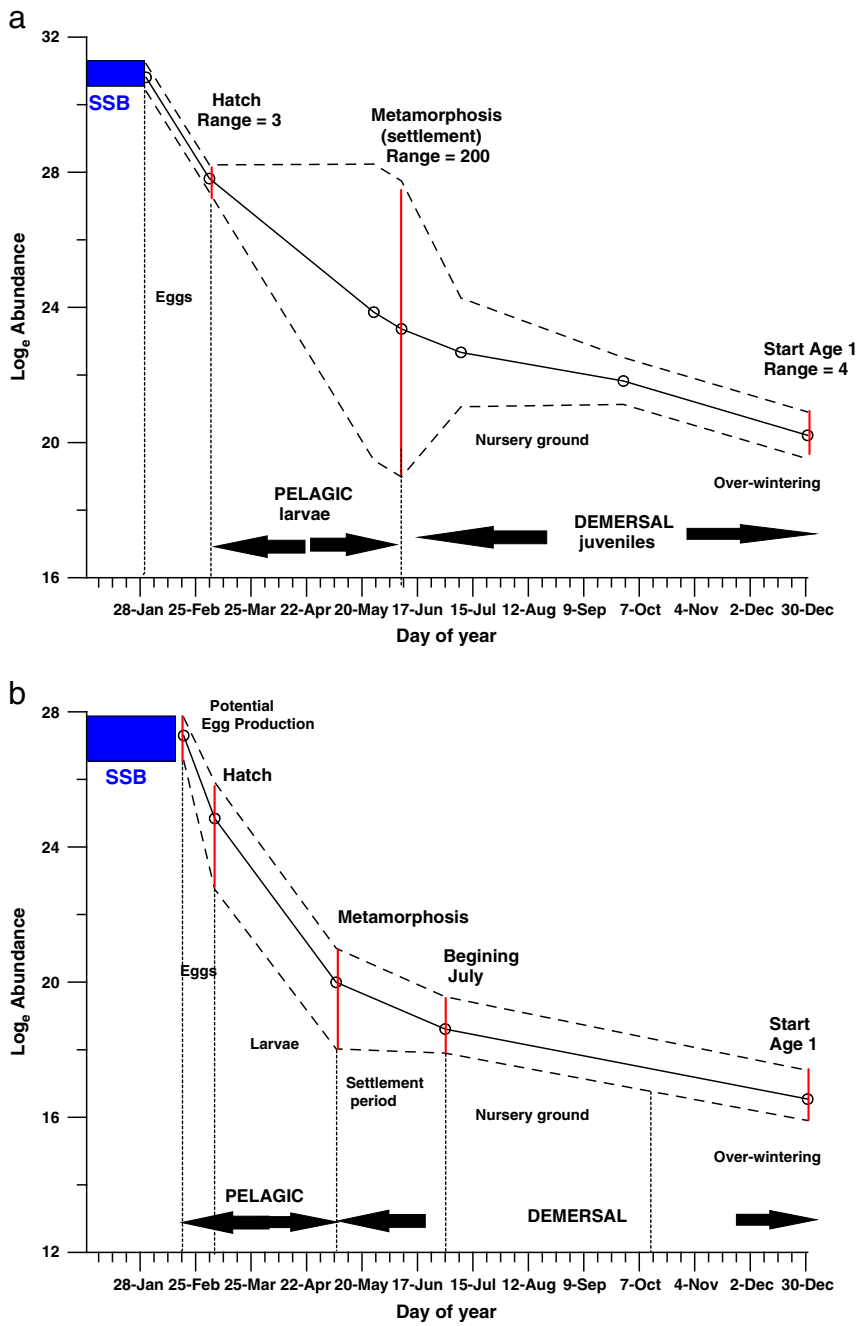

Fig. 3. Abundance (and variability) of early life history stages of European plaice (*Pleuronectes platessa*) in a. the North Sea and b. Irish Sea. North Sea data redrawn from (Beverton, 1995) Irish Sea data from the Paulik diagram (see Fig. 4 and references). Phases are illustrated in Fig. 2. Dashed line indicates the probable range in abundance.

absolute values. The empirical data for the Irish Sea (ICES, 2008) and the Port Erin Bay nursery ground (Nash and Geffen, 1999, 2000, and unpublished updates) suggest that inter-annual variability in survival begins early, based on the relationship between SSB and egg production at the point of stage 1 egg production (Fig. 5, Table 1). In general, larger stage 1 egg productions tend to lead to higher settlement on the nursery ground, however, this is not always the case e.g. 1999 and 2000. Similarly, higher settlement abundances tend to produce larger maximum nursery ground populations, but nursery ground processes probably alter this on an annual basis. Although larger adult populations (as SSB) tend to produce larger populations on the nursery grounds, annually varying processes occurring through the early life history provide the basis for 'recruitment variability'.

3. Pelagic stage mortality

The causes of early life mortality are very diverse with different agents prominent during each life history stage. The pelagic phase is the 'supply side' of the nursery ground phase and factors occurring during the egg and larvae phase determine the numbers of individuals that initially arrive at settlement and then populate the nursery grounds. Flatfish nursery grounds are annually populated with juvenile fish from the new year-class and the nursery ground phase can be broadly characterized as having three phases: the settlement period, summer to autumn growth period, and the emigration period.

Mortality can be considered as a loss from the population whether it is by death, physical displacement out of the system completely, or

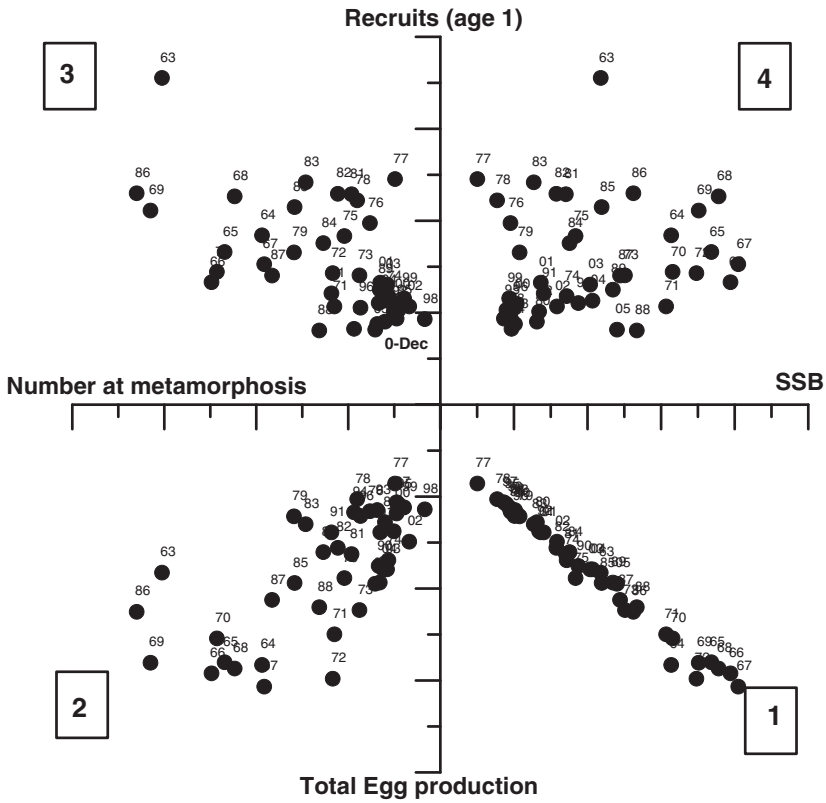

Fig. 4. Paulik diagram for Irish Sea plaice (1963 to 2006 year classes). Updated Fig. 4 from (Nash, 1998). The four panels represent 1. the conversion of Spawning Stock Biomass (SSB) to egg production, 2. the pelagic phase (egg production to metamorphosis or settlement) (bottom left quadrant), 3. the nursery ground phase (settlement to the end of the first winter, age 1) (top left quadrant) and lastly 4. (top right quadrant) the summation of the previous three segments illustrating survival from SSB to the end of nursery phase (in this case 'recruitment').

displacement to an unsuitable habitat resulting ultimately in death or an inability to complete the life cycle. For example, plaice eggs are spawned at certain locations and the eggs and larvae drift downstream toward the nursery areas (Cushing, 1990; Rijnsdorp et al., 1995). Larvae develop, and start to undertake vertical migrations which result in movement inshore to the nursery grounds (Jager, 2001, 2002). However, models suggest that not all larvae reach a nursery area and there are inter-annual variations in their final location (Bolle et al., 2009; Fox et al., 2006, 2009; van der Veer et al., 1998). Losses to unsuitable habitats have not been quantified so there are no empirical estimates of mortality due to these causes.

Table 1
Coefficients of Variation (CV) and ratio between the maximum and minimum for the 1964 to 2006 stock and year classes in the Irish Sea, using the Paulik diagram calculations and CVs through the life tables: using the Irish Sea data with a combination of assessment data and Port Erin Bay (Isle of Man, Irish Sea) for the 1990 to 2003 year classes and stock.

	Paulik Diagram 1964–2006		Assessment and Port Erin Bay 1990–2003
	CV	Ratio (max to min)	CV
Irish Sea SSB	0.33	3.4	0.17
TEP (whole Irish Sea stock)	0.36	3.6	0.33
Stage I egg production (west Isle of Man)			0.38
Numbers of individuals at hatch	0.72	23.8	
Numbers of individuals at metamorphosis	0.67	19.6	
Settlement index (Port Erin Bay)			0.64
Numbers of individuals at beginning of July		4.4	
July population size (Port Erin Bay)			1.08
Maximum population size (Port Erin Bay)			1.03
Numbers of individuals at start of age 1	0.39	4.4	0.14

Egg mortality can be due to endogenous factors such as malformations (see Bunn et al., 2000; Geffen and Nash, 2011, for reviews), and more general factors under the heading of 'egg quality' (Kjorsvik et al., 1990). The extent of this "developmental" mortality has rarely been quantified in the field. Lethal malformations may also be attributed to disease or pollution (Cameron and von Westernhagen, 1997; Cameron et al., 1992; Dethlefsen et al., 1996) or even UV radiation (Pommeranz, 1974; Steeger et al., 1999). These causes also lead to mortality in the larval phase. It is very difficult to assign mortality rates in the field specifically to pollution (Heath, 1992). However, if the final fate of eggs or larvae that are malformed or disturbed by mechanical stress or pollution is an increased vulnerability to predators, their deaths can be considered as part of the predation mortality.

Predation is often considered the main cause of mortality through the egg and larvae stages. There have been a number of estimates of egg mortality rates of plaice, usually associated with estimates of egg productions. In the North Sea instantaneous mortality rates have ranged from 0.02 to 0.17 d^{-1} and varied over years and between areas (Coombs et al., 1990; Harding et al., 1978; Heesen and Rijnsdorp, 1989; van der Land, 1991). Higher mortality rates (0.148–0.291) were noted in the Irish Sea for egg sub-cohorts (Dickey-Collas et al., 2003). The mortality rates were correlated with temperature but no mechanism was identified. If temperature is a mortality factor, this is consistent with the lower mortality in the North Sea compared to the Irish Sea (Table 2) which is generally warmer.

Plaice eggs are relatively large and may experience lower mortality than other species, especially since they are spawned early in the year when there are few invertebrate predators large enough to predate on the eggs. As the zooplankton community develops and grows through the season, the numbers and density of potential predators increase (Bunn et al., 2000). Since egg size declines over the season for plaice (Fox et al., 2003), as for other species (Kjesbu et al., 1991;

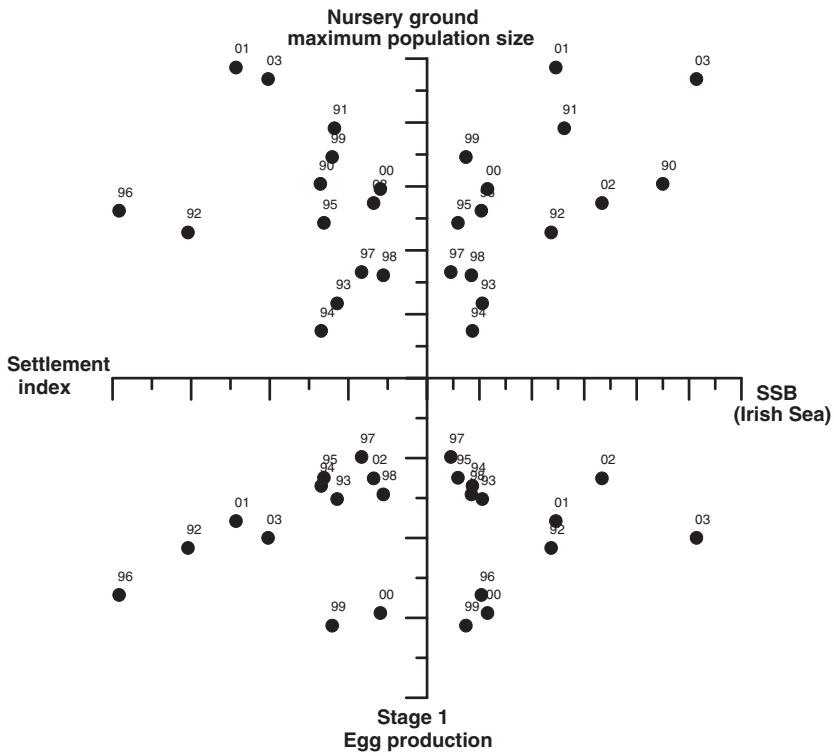

Fig. 5. Paulik diagram for Irish Sea plaice utilizing data from west of the Isle of Man and Port Erin Bay to highlight early pelagic phase processes (stage I eggs in the plankton) and nursery ground processes (settlement and maximum population sizes). See Fig. 4 for explanation of quadrants. Data from Nash and Geffen (1999, 2000), and unpublished data.

Rijnsdorp and Jaworski, 1990), the predation risk increases throughout the spring. This may be partially counteracted by size-specific egg development rates (Fox et al., 2003). Large year classes are often associated with cold winters. Lower temperatures may reduce predator abundance, or alter their distribution patterns and feeding rates. In addition reduced temperatures will delay egg development and hatching, which could put feeding larvae in better synchrony with their prey (e.g. Cushing's match–mismatch theory — (Cushing, 1975, 1990)). Even if the predation on the larvae was not reduced the better feeding conditions could have enhanced larvae survival.

Daan et al. (1985) estimated that North Sea herring consumed up to 1.9% of the eggs produced by plaice. The majority of the eggs were consumed by young herring in the vicinity of the plaice spawning grounds. Ellis and Nash (1997) found plaice eggs in the stomachs of sprat and herring in the Irish Sea, and Segers et al. (2007) in the North Sea, with late stage eggs selected, presumably because they were more pigmented and easier to see. The evidence for stage dependent mortality rates argues against the use of fixed mortality rates. This is especially important in some procedures such as the Annual Egg Production Methods (AEPM) which rely on mortality rates calculated from the decline in egg number with age in order to back calculate the numbers of eggs spawned (Armstrong et al., 2001). If the mortality rate is different between spawning and stage 1, a period when eggs can be very concentrated and when they are less visible due to a lack of pigment, then the estimate of how many eggs were spawned can be wildly inaccurate (Bunn et al., 2000). Gelatinous zooplankton predators may also exert a significant predatory pressure on the egg and larvae stages but there are few quantitative measures of the mortality rates (Bailey and Batty, 1984; Bax, 1998; Lynam et al., 2005; van der Veer, 1985).

Predation has been notoriously difficult to find and to measure, especially on larvae. Bannister et al. (1974) estimated larval plaice mortality rates in the North Sea ranging from 0.0095 to 0.0611 d^{-1} with lower mortalities in colder years, despite lower growth rates and longer development times.

Table 2
Comparison between the average instantaneous mortality rate in the Irish Sea with the rates for the North Sea reported by Beverton and Iles (1992a,b).

Life history stage	Instantaneous mortality rate d^{-1}	
	Irish Sea	North Sea (1986–89 year classes)
Egg stage	0.157	0.094
Larvae stage	0.079	0.046
During the settlement period	0.025	0.023
Nursery ground phase	0.011	0.014

4. Settlement mortality

There are few calculations of mortality rates for young flatfish during the settlement period. This is generally because the standard calculation methods rely on estimating the reduction in numbers over time, and during this period fish are arriving on the nursery ground whilst those already there are dying. Zijlstra et al. (1982) and van der Veer (1986) estimated early mortality rates of plaice settling in the Wadden Sea, prior to mid June. Mortality rates in selected size classes over relatively short periods of time suggested density dependent mortality, reaching a maximum of approximately 0.08 d^{-1} (Beverton and Iles, 1992a; van der Veer, 1986). Nash and Geffen (2000) used repeated, regular sampling of settling fish in Port Erin Bay, Irish Sea to estimate abundance, and over the period 1991 to 1998 the 'settlement period instantaneous mortality' ranged from 0.017 to 0.086 d^{-1}. It is often stated that the pelagic phase ends when larvae metamorphose and settle. However, metamorphosis and settlement are not necessarily coupled (Geffen et al., 2007). In

flatfishes specifically, the term metamorphosis is used to define morphological (the most radical involving eye migration, skull torsion, and a change in shape with lateral flattening of the body) and physiological development. The term settlement defines behavioral and ecological changes associated with the transition of larvae from the 3-dimensional planktonic environment to the 2-dimentional demersal way of life (Geffen et al., 2007; Gibson and Batty, 1990). In plaice populations, both the pre-metamorphosed larvae and metamorphosed individuals are found in the vicinity of the nursery grounds. Very often the newly metamorphosed individuals have not truly settled on the bottom, still spending considerable amount of time feeding in the water column. Therefore, settlement is not instantaneous and young flatfish could be vulnerable to predation with their shape, sensoria, and coloration awkward for a midwater existence. The shift from pelagic to demersal life also involves a shift in diet due to different prey being available and predation techniques due to the change in prey types and their behavior and their habitat.

Gibson and Batty (1990) examined the effect of substratum type on growth of late larvae and timing to metamorphosis for plaice in aquaria. Although plaice cannot delay metamorphosis in order to find suitable habitats, individuals that are over unsuitable substrata tend not to feed and keep moving. In this indirect way, the substratum may impair growth and affect the timing of metamorphosis. Although late stage plaice larvae can live for many days without feeding (Wyatt, 1972) while searching for suitable habitat, there comes a point when the larvae will die of starvation or be incapable of avoiding predation. Experimental studies also showed that settling plaice spent less time on the bottom where appropriate food was limited and where known predators occurred (Wennhage and Gibson, 1998).

In some nursery areas, especially where tidal currents are particularly influential, settlement or influxes to the nursery area can be periodic (Rauck, 1974), in 'waves' linked to spring or neap tides. These sudden increases in density may lead to periodic predator satiation, although this aspect of settlement mortality has been little studied. The periodic settlement pattern results in nursery ground sub-cohorts and age structuring within the juvenile population (Alhossaini and Pitcher, 1988; Al-Hossaini et al., 1989; Dau, 1994; Geffen et al., 2011).

Mortality is generally high at settling and soon after (Wennhage, 2002) since the juvenile plaice are naive to the benthic environment as they switch from a pelagic to a demersal way of life. Their defense mechanisms e.g. burying are not fully developed at first, but the ability to avoid predation rapidly increases after metamorphosis (Wennhage, 2000). Brown shrimp (*Crangon crangon*) and shore crabs (*Carcinus maenas)* have been identified as major resident predators on young plaice (Ansell et al., 1999; Edwards and Steele, 1968; Oh et al., 2001; van der Veer and Bergman, 1987; Wennhage, 2002). In the Wadden Sea, heavy predation by *Pleurobrachia pileus* (van der Veer, 1985) and brown shrimp (van der Veer and Bergman, 1987) can remove a large proportion of the pre-settling and settling individuals. Analyses of shrimp stomach contents indicate that predation is largely density-dependent and concentrated on newly-settled plaice (12–17 mm), decreasing rapidly for larger fish (Burrows et al., 2001; Gibson et al., 1995; van der Veer and Bergman, 1987; Wennhage, 2000; Wennhage and Pihl, 2001). One interesting variation occurs with a link between settlement size and/or subsequent growth rates whereby young stages can spend variable amounts of time in a potential predation window. Whilst the instantaneous mortality is not altered, the 'stage duration' varies and thus the absolute mortality or loss of individuals changes (Witting and Able, 1993). The potential predation pressure by larger fish or even birds during settlement has not been investigated in plaice. However, restocking programs for other flatfish species indicate that predation by birds in some areas may be quite high (Leopold et al., 1998; Sparrevohn and Støttrup, 2007).

5. Nursery ground mortality

5.1. Overall nursery ground population dynamics

The mortality rates for the nursery ground phase are generally estimated from the reduction in numbers after the completion of settlement, as summarized by Iles and Beverton (1991). Plaice nursery ground mortality is primarily density-dependent, but there is also a level of mortality that is not related to fish density. The empirical data suggests a 'base' level of mortality which only starts to increase above a threshold density. Lockwood (1980), using data from a number of UK plaice nursery grounds (Firemore Bay, Loch Ewe; Filey Bay, Yorkshire; Port Erin Bay, Isle of Man; Red Wharf Bay, Angelsey) suggested that the density independent mortality rate was between 0.01 and 0.021 d^{-1}. Beverton and Iles (1992a) used information for a range of flatfish nursery ground populations and suggested much lower values, up to 0.01 with a general threshold for density-dependent processes of 0.18 fish m^{-2}. The level where density dependent factors begin to take effect depends on the numbers of individuals relative to the productivity of the nursery.

Beverton and Iles (1992a) modeled the density dependent component of nursery ground mortality for the period July–January for juvenile flatfish nursery grounds as $M = 0.01 + 0.0044 \times \log_e$ (Maximum Density). This relationship indicates the level of mortality to be expected on nursery grounds based on population density, but not the mechanisms that are operating or the dynamics of the nursery grounds.

Density-dependent factors are also important in interpreting the relationship between mortality rate (M) and growth rate (G), expressed as M/G or physiological mortality (Beyer, 1989; Houde, 1997, 2008, 2009) which can be considered an index of cohort specific survival potential and responsiveness to environmental or biological processes. Low values of M/G in early life, a fast decline in value, and transition to values of $M/G < 1$ at young ages, are generally associated with high recruitment. In fact, Cushing (1975) proposed that variation and value of the inverse ratio i.e. growth to mortality (G/M) could be used as an index of the potential for the size of a recruiting year class. An alternative view of the relationship between M and G relates to habitat suitability (Bohlin et al., 1994; Werner and Gilliam, 1984). For example, the steep relationship between mortality and growth ($-M/G$) as seen by Nash et al. (2007) for plaice in Port Erin Bay in 1994 and 1998 and Ardmucknish Bay in 1996 and 1997 (Table 3), indicates a greater pressure to emigrate off the nursery ground. However, predation pressure is likely to be higher offshore (Gibson et al., 2002; Ryer et al., 2010) so the survival potential can only increase if there are also more opportunities for increased growth rate offshore. Bohlin et al. (1994) suggested that ontogenetic habitat shifts which minimize M/G are more likely to occur for segments of a population that are not regulated by density-dependent growth and/or mortality.

Density-dependent effects on mortality and growth rates are often difficult to discern because of masking factors such as temperature effects. This is the case where cold winters tend to produce large year classes and elevated densities which should result in low growth rates. However, reduced growth rates are also a direct result of the cooler water temperatures, and so the ultimate mechanism that links forcing factor to outcome is obscured. Density-dependent processes imply limited resources, competition and consideration of carrying-capacity (Kashiwai, 1995), defined as the capacity of an ecosystem to support a population. This carrying capacity is dynamic and changes with the annual productivity. In addition there is interplay between numbers of individuals and size of individual in how these make up the total biomass that can be supported within an ecosystem. There can be many small individuals or fewer large ones, as expressed by the concept of dynamic thinning lines (Begon et al., 2006) and the relationship between mean weight of individuals and

Table 3
Mean water temperature °C (1st May–31st August), and daily instantaneous growth rate (g) (May to September), instantaneous early (EMR) and standard (SMR) mortality rate (d^{-1}) (see Nash and Geffen, 2000) and ratio of mortality rate to growth rate (−m/g) of juvenile plaice in Ardmucknish (Oban) and Port Erin (Isle of Man) Bays (data updates from Nash et al., 2007).

Year	Ardmucknish Bay				Port Erin Bay				
	Mean water temperature (°C)	Inst. Growth rate (g)	Inst. mort. (SMR) (m)	−m/g	Mean water temperature (°C)	Early mort. (EMR)	Inst. Growth rate (g)	Inst. mort. (SMR) (m)	−m/g
1963*					11.04			0.0161	
1964*					12.15			0.0132	
1965*					12.04			0.0235	
1986	10.67	0.013	0.0088	−0.67					
1987	10.67	0.020	0.0036	−0.18					
1988	11.92	0.016	0.0159	−1.00					
1989	12.60	0.011	0.0101	−0.92					
1991					11.70	0.0318		0.0123	
1992					12.36	0.0770		0.0108	
1993					11.86	0.0248		0.0160	
1994					11.69	0.0309		0.0238	
1995	12.03	0.013	0.0115	−0.86	12.58	0.0207	0.015	0.0290	−1.96
1996	11.85	0.014	0.0154	−1.10	11.70	0.0862	0.015	0.0212	−1.45
1997	12.13	0.015	0.0180	−1.18	12.43	0.0234	0.014	0.0100	−0.69
1998					12.91	0.0145	0.012	0.0160	−1.31
1999					12.87	0.0624	0.015	0.0141	−0.93
2000					12.84	0.0320	0.042	0.0299	−0.71
2001					12.68	0.0865			
2002					12.88	0.0058	0.011	0.0227	−1.98
2003					13.29	0.0094	0.026	0.0162	−0.63

* Data from Iles and Beverton (1991).

density (Dunham et al., 2000). Nash et al. (2007) applied the dynamic thinning line concept to juvenile plaice nursery grounds, and demonstrated that for a nursery at 'carrying capacity' as the fish increased in weight over time, their density declined (Fig. 6). However, the speed at which mean weight increases is a function of how rapidly the density is reduced (the ambient mortality rate). Self thinning lines indicate the level of the carrying capacity and how the total biomass can be distributed across a combination of mean weight and density. It also shows how the two vary in tandem. If low spawning stock sizes produce fewer settling plaice, the numbers of juveniles on the nursery grounds will rarely reach the carrying capacity and thus rarely invoke any density regulating effects (Keeley, 2003).

Reductions in growth rate of juvenile plaice on nursery grounds after August were noted from tagging studies by Nash et al. (1994). More recent work by Teal et al. (2008) in the Wadden Sea also noted reduction in growth rate late in the summer (July to September) and suggest this was due to food limitation. The inclusion of examinations of growth rates using Dynamic Energy Budgets (DEB), as reported by van der Veer et al. (2010), again for the Wadden Sea, suggests that growth was less than the predicted maximum in the summer months but there was no plaice density caused negative effects on growth rate. They do suggest that there is interspecific competition for food i.e. epibenthic species are competing for food and that both food quality and quantity affect the growth rates. How this directly impacts mortality rates is uncertain, however, reductions in growth rates could increase the time individuals remain within a predation 'window'.

Most authors attribute the population regulating mortality in plaice to predation (Bax, 1998). As well as invertebrate predators there are also a number of fish species that eat juvenile plaice (Ellis and Gibson, 1995; van der Veer et al., 1997). Some juvenile plaice remain on the nursery ground to overwinter there (Nash et al., 1994) or return as 1+ or older juveniles and cannibalize the O-group population (Edwards and Steele, 1968; Lockwood, 1980; Macer, 1967; Riley and Corlett, 1966). As the season progresses and the fish become larger there is probably an increase in the avian predation, for example by cormorants (Leopold et al., 1998). However, little quantitative research has been undertaken on this source of mortality.

5.2. Population structure and sub-cohorts

Over the course of time the size distribution of a population will develop and change due to a combination of individual growth and mortality rates. Mortality is not evenly distributed across the populations and this can produce varying inequalities in size distribution (as measured by the Gini Coefficient (Nash et al., 2007)). It can be difficult to detect size selective mortality because the mean lengths of the nursery ground population are dynamic, both annually and inter-annually, and generally increase over the season. Using the Centre of Gravity (CG), which represents the relative position in time of a population's maximum diversity in size (adapted from Gould et al. (1987)), Nash et al. (2007) demonstrated that size-selective processes differed between years (Fig. 7), and in general higher instantaneous mortality rates result in lower CGs. Low CGs suggest that higher mortality rates operating on a size selective basis have made the population more uniform in size at an earlier point in the season.

An important structural element of juvenile plaice populations, especially those on the west coast of the British Isles, is the presence of sub-cohorts (Al-Hossaini et al., 1989; Allen et al., 2008; Geffen et al., 2011). These sub-cohorts are generated at settlement, and are caused by periodicity in the spawning or hatching (Allen et al., 2008), aggregations synchronized to tidal cycles (Geffen et al., 2011; Rauck, 1974), or consecutive waves of incoming settlers originating from different spawning locations (Al-Hossaini et al., 1989). The studies in the Irish Sea have generally identified three main sub-cohorts with a group of later settlers trickling in over time (see Al-Hossaini et al., 1989; Geffen et al., 2011). There are different mortality rates associated with the sub-cohorts. Geffen et al. (2011) noted that the first sub-cohort is usually the largest and suffers the highest mortality. Very large year classes may occur through a release of predation pressure on this first sub-cohort. The second sub-cohort generally has the lowest mortality of the three. Since many of the nursery ground predators are size selective (Ellis and Gibson, 1995; Sogard, 1997), the presence of these sub-cohorts with different lengths and different growth rates adds to the complexity of the fine scale dynamics of these nursery areas.

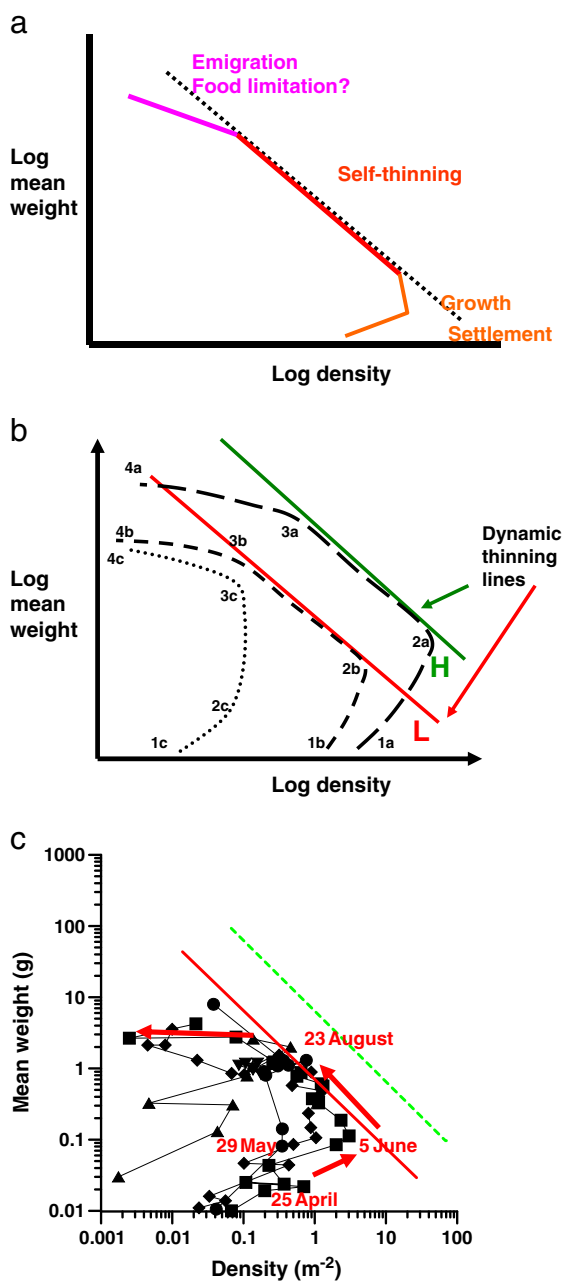

Fig. 6. Illustrations of dynamic thinning lines on juvenile plaice nursery grounds. a. Theoretical model illustrating three phases within dynamic thinning lines. b. the location of dynamic thinning lines for high (H) and low (L) productivity nursery grounds. Dashed lines a, b and c illustrate the trajectories for 1a–4a high productivity, 1b–4b low productivity and 1c–4c low settlement periods. c. examples of trajectories for Port Erin Bay and dynamic thinning lines (solid = Port Erin Bay, Isle of Man; dashed = Ardmucknish Bay, Lynn of Lorne, Scotland), redrawn from Nash et al. (2007).

Spatial differences in growth rates, such as those observed in Red Wharf Bay (Ciotti et al., 2010), will result in spatial differences in population size structure since both Riley (1973) and Burrows et al. (2004) have clearly shown that young plaice have a tendency for site or location fidelity. In both cases they demonstrated that displaced juvenile plaice return rapidly to their preferred locations and there is very little along shore displacement of individuals.

Tidal factors are important in juvenile plaice nursery grounds, and there is an energetic cost for young plaice to make their extensive tidal migrations (Gibson, 1973), which can be as long as 3–4 km over the mudflats at Morecambe Bay (eastern Irish Sea). To remain stationary in these areas the young fish run the risk of being 'dried'

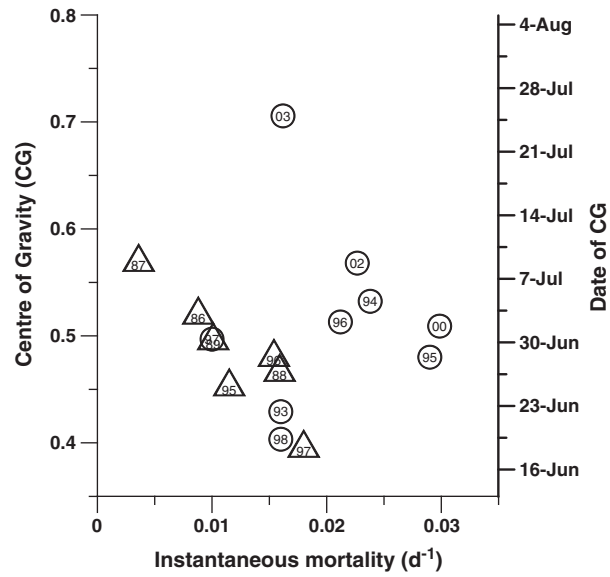

Fig. 7. Relationships between instantaneous mortality rates and the size structuring (as measured by the Centre of Gravity (Gould et al., 1987; Nash et al., 2007)) of juvenile plaice on nursery grounds. The date of maximum diversity in length is given on the right hand axis.

out as the water recedes or being in deep water where the predation risk may be higher (Gibson et al., 2002). Young plaice may be more vulnerable to predators between dusk and dawn, so tend to concentrate in shallower water at night (Burrows, 2001). Balancing tidal migration with predator avoidance and foraging behavior probably leads to an uneven distribution of mortality over time on the nursery grounds. The weekly disruptions to feeding for juvenile plaice in Port Erin Bay can been seen in a pattern of check marks on the otoliths, which are synchronous with a combination of tidal and light (day: night) cycles (Geffen and Nash, 1995).

5.3. Apparent mortality

Most calculations of mortality rate are based on observed reductions in numbers, but there are other factors that can confound the estimates for juvenile flatfish. Firstly the sampling equipment (generally beam trawls for juvenile plaice) is size selective and size dependent efficiency factors need to be applied to estimate the true population size and size distribution (Edwards and Steele, 1968; Kuipers, 1975; Kuipers et al., 1992). More importantly later in the season, especially with small trawls and small mesh size, there can be an increasingly greater amount of gear avoidance by larger fish. The reduction in catches can be exacerbated by the size related emigration off the nursery grounds in to deeper water (Gibson et al., 2002; Nash et al., 2007). This is usually countered by defining the nursery area, ensuring it is sampled adequately and being aware of when fish are likely to be leaving the nursery area. Abundance data from the period of emigration should not be used in the estimate of mortality.

6. Conclusions

The early life history dynamics of European plaice have been well studied and the levels of mortality through this period are generally known. There are differences in mortality rates between stocks and this probably reflects the productivity of the ecosystems and their ecology. The use of shallow water nursery areas tends to concentrate the post-settlement juveniles in well defined demersal areas (Beverton, 1995). The shallow water and proximity to coastlines have allowed a considerable body of research to be built up

covering the dynamics of these nursery areas. New immunological techniques are now available to allow more detailed analyses of predator–prey relationship, especially the role of juvenile flatfish in the diets of invertebrate predators (e.g. Albaina et al., 2010; Taylor, 2004). In addition, new modeling approaches are able to elucidate some of the interactions between predators and prey, such as in Burrows et al. (2001) and Dynamic Energy Budget (DEB) modeling (van der Veer et al., 2001).

The extensive time series from studies of plaice nursery grounds have revealed a complex structure to the populations with complex predator–prey and behavioral interactions. These processes affect both intra and inter-annual levels of mortality for juveniles that are often overlooked in estimations of early life mortality rates. Whilst the research is comprehensive it is by no means complete. We may know, or think we know what the mortality rates are but we are still not confident that we know or understand all the processes driving the mortality.

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